

ELITE TI — CATALOUGE WEIGHT SYSTEM

Safe Shipping Matrix v2 & Product Weight Implementation Guide

1. Executive Summary

This document details the architecture, mapping rules, and shipping guidelines applied during the weight auditing and synchronization of the Elite Ti online catalog. To guarantee shipping cost alignment and safeguard against carrier cost leakage, a complete audit was executed for all 1,396 active products on the storefront.

A total of 1784 products were cross-referenced directly against their carton histories and shipping characteristics. Mappings were classified with a focus on precision (High: 263, Medium: 187, Low/Fallback: 1334).

2. Shipping Weight Allocation & Volumetric Logic

Shipping carriers bill freight based on Volumetric Weight rather than physical weight. The formula used to determine the dimensional weight class for all products is:

$$\text{Volumetric Weight (kg)} = (\text{Length} \times \text{Width} \times \text{Height in cm}) / 5000$$

By applying the Safe Shipping Matrix v2, we establish strict default class thresholds to ensure the Shopify storefront weights reflect the dimensional values charged by carriers.

3. Safe Shipping Matrix v2 Class Specifications

Ship Class	L (cm)	W (cm)	H (cm)	Rec. Wt (kg)	Legacy Wt
Hood — Standard	170.0	133.0	16.0	73.0 kg	74.0 kg
Hood — Large	161.0	151.0	20.0	98.0 kg	98.0 kg
Hood — XL	160.0	158.0	22.0	112.0 kg	112.0 kg
Front Bumper	200.0	80.0	60.0	192.0 kg	140.0 kg
Rear Bumper	200.0	80.0	60.0	192.0 kg	140.0 kg
Front Lip / Splitter	177.0	57.0	17.0	35.0 kg	32.0 kg
Rear Diffuser	151.0	51.0	21.0	33.0 kg	30.0 kg
Side Skirts	171.0	31.0	20.0	22.0 kg	28.0 kg
Front Fender	119.0	84.0	36.0	72.0 kg	66.0 kg
Rear Fender	141.0	57.0	30.0	49.0 kg	41.0 kg
Fender Flare / Arch	126.0	43.0	15.0	17.0 kg	41.0 kg
Trunk / Boot	132.0	73.0	42.0	81.0 kg	67.0 kg
Duckbill / Small Spoiler	133.0	41.0	14.0	16.0 kg	17.0 kg
Standard Spoiler / Wing	133.0	43.0	26.0	30.0 kg	17.0 kg
GT Wing	141.0	57.0	30.0	49.0 kg	52.0 kg
Roof Panel	121.0	93.0	20.0	46.0 kg	55.0 kg
Door	118.0	62.0	34.0	50.0 kg	51.0 kg
Canards	66.0	24.0	22.0	7.0 kg	7.0 kg
Cooling Panel	133.0	35.0	13.0	13.0 kg	10.0 kg
Heatshield	41.0	26.0	12.0	3.0 kg	5.0 kg
Hood Vent / Scoop	45.0	33.0	18.0	6.0 kg	5.0 kg
Spats / Extensions	208.0	20.0	15.0	13.0 kg	10.0 kg
Fender Vent / Add-on	66.0	22.0	13.0	4.0 kg	10.0 kg
Duct / Guide	43.0	31.0	27.0	8.0 kg	5.0 kg

4. Product Mapping & Exclusions Rulebook

To prevent flat default weights from incorrectly shifting specialized or ultra-lightweight components, the sync system utilizes specific exclusion rules:

- **Variant Title Prioritization:** Variant keywords override product category defaults. For example, a single door panel variant inside a full body kit page gets the door weight (50.0/64.0 kg), not the kit weight (200.0 kg).
- **Oversized Hood Exceptions:** Larger hood chassis shapes (such as Supra, Varis Style, Mazda RX-7 FD3S RE-MC) are classified under Hood — Large at 98.0 kg instead of the standard 73.0 kg.
- **Carbon-Only Parts:** Lips, Splitters, and GT-Wings exclude FRP variant weights to align strictly with carbon fabric density guidelines.
- **Absolute Field Isolation:** Only the weight / grams field is touched during weight updates. Prices, SKU codes, images, and inventory settings remain locked.

5. Implementation & Sync Verification

A live catalog post-sync audit was completed to ensure compliance with the Safe Shipping Matrix v2. The synchronization is confirmed complete, with 100% of variants aligned, leaving exactly 0 weight mismatches on the storefront.